Lecture IV
Spatial Linear Models

STAT 679: Statistical Methods for Spatial Lattice Data-Spring 2026
University of Wisconsin-Madison

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Outline

1. Fitting Spatial Linear Models by Maximum Likelihood
2. Model Selection
3. R-Demo
4. Assignment

Recap...

- Lecture I: Spatial data, neighbors, and join tests
- Lecture II: Spatial correlation (Moran's I and Geary's c)
- Lecture III: Ways to model spatial dependence in the error for valid statistical inference. (namely, SAR, CAR, MA)

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Model Fitting via Maximum Likelihood

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- Consider a linear regression model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i$$

where $\varepsilon_i \sim$ iid $N(0, \sigma^2)$.

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- Consider errors ε_i that follow SAR(1) with

$$\begin{aligned}\Sigma &= \sigma^2 \left[(I - \rho N)^T D^{-1} (I - \rho N) \right]^{-1} \\ &= \sigma^2 (I - \rho N)^{-1} D (I - \rho N^T)^{-1}\end{aligned}$$

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  or MA(1) with

$$\Sigma = \sigma^2 (I + \rho N) D (I + \rho N)^T.$$

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- The residuals are not the usual ones. R calculates the residuals such that the residuals are independent and have constant variance. Hence Moran's I can be used to check spatial dependence among the residuals.

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- Consider spatial linear models with **SAR(1)** errors:

$$\text{(I)}: \quad y = \beta_0 + \beta_1 x_1 + \beta_3 x_3 + \beta_6 x_6 + \varepsilon, \qquad \ell = -472.536$$

$$\text{(II)}: \quad y = \beta_0 + \beta_2 x_2 + \beta_4 x_4 + \beta_5 x_5 + \beta_7 x_7 + \beta_8 x_8 + \varepsilon, \qquad \ell = -432.260$$

$$\text{(III)}: \quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \varepsilon, \qquad \ell = -431.144$$

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- However, **it is not necessarily the case** that the **model with the larger loglikelihood value is the better model**.
- If two **models are nested, consider performing likelihood ratio** test. Otherwise use an information criterion such as **AIC or BIC**.

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- In the WI & IL poverty example, consider

$$H_0 : [\beta_1 = 0, \beta_3 = 0, \text{ and } \beta_6 = 0 \mid \beta_0, \beta_2, \beta_4, \beta_5, \beta_7, \beta_8]$$

versus

$$H_A : [\beta_1 \neq 0, \beta_3 \neq 0, \text{ and } \beta_6 \neq 0 \mid \beta_0, \beta_2, \beta_4, \beta_5, \beta_7, \beta_8].$$

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where the df for χ^2 is the number of parameters being tested.

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- In the WI & IL poverty example,

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- Compare with χ^2_3 and p-value $= 0.526$. We do not reject the null hypothesis that $\beta_1 = 0$, $\beta_3 = 0$, and $\beta_6 = 0$.

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- The likelihood ratio test works when the reduced model is a proper subset of the full model, but does not work for comparing two models that are not nested within each other (e.g. model (I) versus model (III)).

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- Caution: Comparing models with AIC does not work between transformed and untransformed response data. That is, we cannot compare

$$\ln(y) = \beta_0 + \beta_1 x + \varepsilon \qquad \text{versus} \qquad y = \beta_0 + \beta_2 x_2 + \varepsilon.$$

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